Experiment No. 3

Performance Comparison Between Three-Phase Uncontrolled and Controlled Three Phase Rectifier

Objectives:

1. To analyze six pulse uncontrolled and controlled three phase rectifiers.
2. To observe the improvement in the quality of output DC voltage
3. To observe decrease in ripple factor as compared to single phase half and full wave rectifiers.

Section 1:

Arrange the circuit as shown in Figure 1 and energize it using a three-phase voltage of $230V_{LN,RMS}$ and 50Hz. Don't forget to have a phase difference of 120 degrees between the phases. Consider an RL load of 10 Ω and 50 mH for your analysis. Do calculations about the peak, average and RMS current of your diode such that it can feed your load. Search the diode which can meet your specifications online, from vendors like mouser.com or digikey.com. Now make sure you can get its spice model online, please import it inside LTspice and run your simulation.

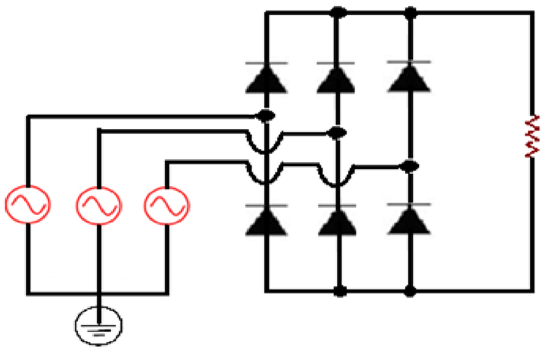
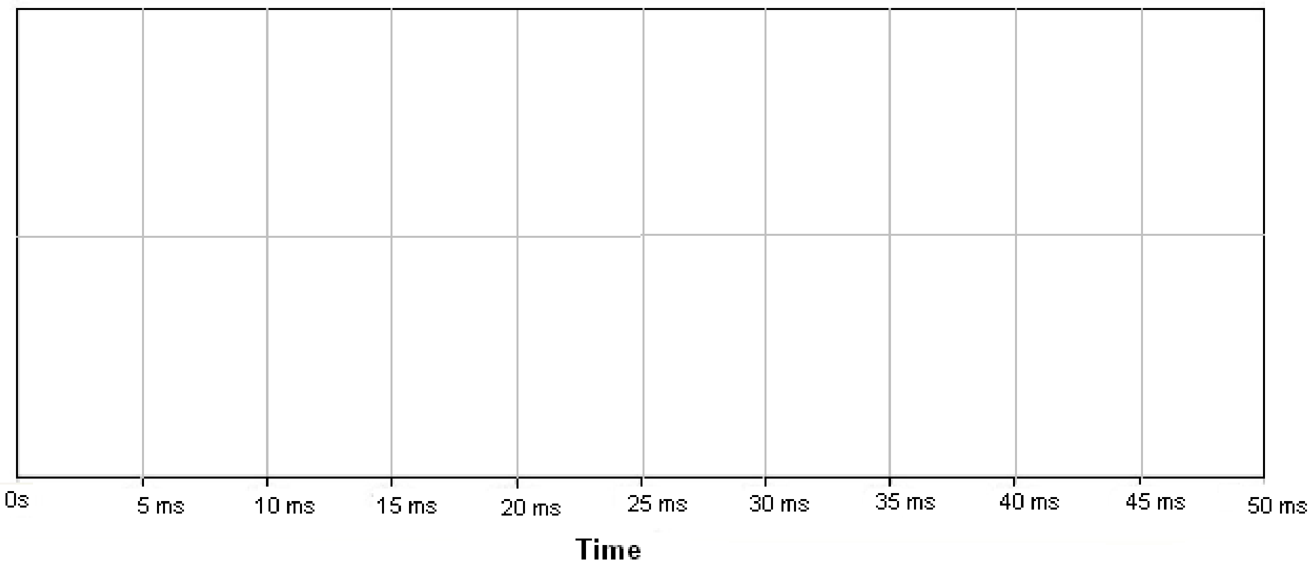


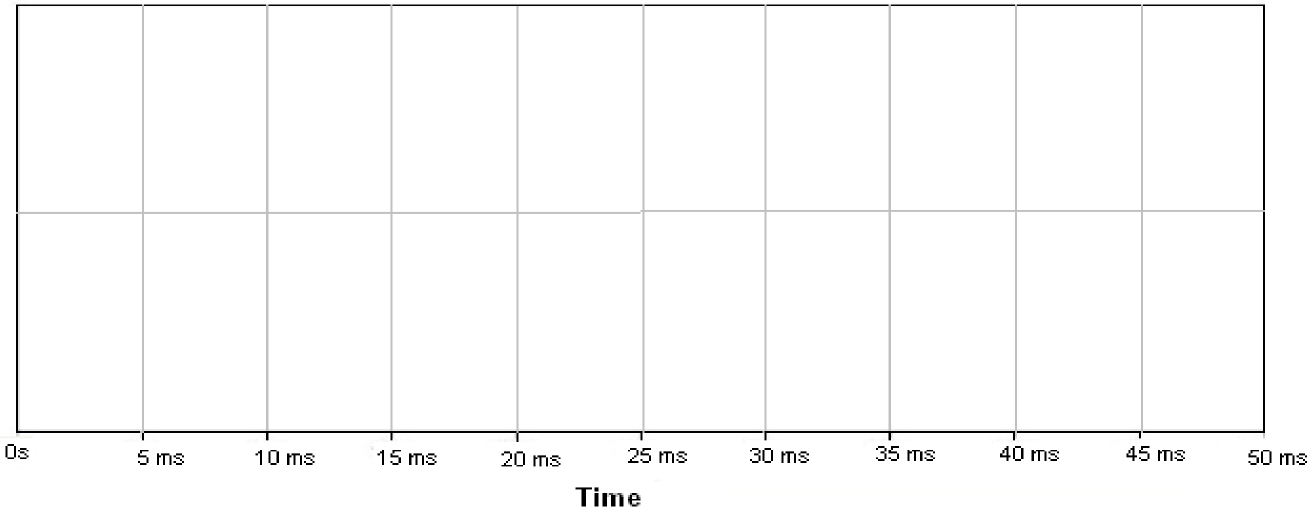
Figure 1

Observe and plot the following waveforms

1- **Sketch voltage across load (resistor inductor).**



2- Sketch current through the diodes of phase A. Also sketch current of phase A of input power supply.



<u>Section 2:</u>

Design an output capacitor using the concepts taught in today's class. Consider ripple factor of 20% for your calculations. Also remember that each frequency cycle 'f' will be split into '6' equal pulses, so you can take 'f/6' in your calculations. Redraw the above asked quantities one more time.

<u>Section 3:</u>

Replace the diode with a SCR from vendor of your choice and repeat section 1 and section 2 for a firing angle of 30 degrees and 45 degrees. You can generate the pulses for SCR using a voltage source and controlling its pulse duration and pulse width. Please don't forget to locate the SCR with the right characteristics.